

Game 11

Welcome to the Avondale DSS Model

Notes:

There are two operational sheets to this model

- Data Model** This describes the previous year's clinical and cost activity
The key cost coefficients can be extracted from this data
- Cost Model** This is the main investigation model.
This model allows you to make adjustments to the various inputs
and calculates what the expected costs will be

Colour Key:

- black values - are for pure data
- blue values - are for calculations/formulas and should not be altered
- grey values - are for intermediate calculations/formulas
- yellow cells** - are input cells, and can be adjusted by the user

Avondale - last year's activity

	Ward	Episodes	Beds	Fixed Cost Weight	Total Cost	Fixed	Variable
GM	1	996	25	0.7	1,000,000	700000	300000
GM	2	895	25	0.7	950,000	665000	285000
GM	3	1194	25	0.7	1,050,000	735000	315000
GM	4	1000	25	0.7	1,025,000	717500	307500
GM	5	915	25	0.7	975,000	682500	292500
Σ		5,000	125		5,000,000	3,500,000	1,500,000

GS	8	874	20	0.7	955,000	668500	286500
GS	9	865	20	0.7	1,044,000	730800	313200
GS	10	915	22	0.7	1,113,000	779100	333900
GS	11	975	24	0.7	1,177,500	824250	353250
GS	12	971	24	0.7	1,210,500	847350	363150
Σ		4,600	110		5,500,000	3,850,000	1,650,000

Required outputs:

	GM	GS
Specialty Fixed Costs:	£3,500,000	£3,850,000
(per ward)	£700,000	£770,000
Overhead Costs:	£1,000,000	£940,000
Variable Cost per Episode:	£300.00	£358.70
Average LOS:	7.5	7.0
Total Episodes:	5,000	4,600

notes:

(total for specialty)
(average by ward)
(total for specialty)
(ave. for one episode)
(in days)
(all patients)

	GM	GS
Overall Cost per Episode:	£1,200.00	£1,400.00
Total Hospital Cost:	£6,000,000	£6,440,000
Σ :	£12,440,000	

Key:

Black is data
Blue is a formula

General Medicine

National GM incidence rate	0.0464	per person
district population	158,238	<i>last census</i>
District weighted pop ² (GM)	165,025	

Specialty Reference Costs

(based on current number of episodes)	
For whole District	£1,340.18
At Avondale	£1,279.83
At Bridge	£1,471.37

General Surgery

National GS incidence rate	0.0474	per person
district population	158,238	
District weighted pop (GS)	172,987	

Specialty Reference Costs

(based on current number of episodes)	
For whole District	£1,336.64
At Avondale	£1,430.26
At Bridge	£1,186.85

Class	Ref cost ¹ (£)	Avondale incidents ³	Bridge incidents ³	total district incidents	National incidence rates (per 1000 weighted persons)
1 Epilepsy	1036	274	228	502	3.68
2 Asthma and Respiratory	1150	404	192	596	4.24
3 Pleural	802	566	34	600	4.28
4 Heart and Circulation	2160	498	700	1198	6.89
5 Cerebral	1389	658	250	908	6.1
6 Fluid Balance	1260	469	208	677	4.42
7 Pain and Palliative	1023	391	62	453	3.31
8 Gastroenterology	1154	538	246	784	4.89
9 Metabolic	820	463	235	698	3.21
10 Diabetic	1645	739	145	884	5.54
Σ		5000	2300	7300	

Class	Ref cost (£)	Avondale incidents ³	Bridge incidents ³	total district incidents	National incidence rates (per 1000 weighted persons)
1 Oesophagal	946	177	489	666	4.39
2 Large Intestine	1754	1183	142	1325	6.48
3 Small Intestine	1203	282	356	638	4.14
4 Stomach & Duodenum	1623	829	227	1056	5.8
5 Rectal	1280	542	467	1009	6.17
6 Lungs	1428	495	300	795	5.041
7 Inflammatory Disorders	894	254	577	831	3.648
8 Bowels	1117	267	256	523	3.151
9 Hernias	1375	634	96	730	4.59
10 Biliary Tract	1025	137	90	227	1.595
Σ		4800	3000	7800	

Notes

1. The reference cost is an average price taken across many hospitals and is thus a guide price for CCGs to use in their contracting. Hospitals often don't account for costs in this way though and only calculate them at specialty level, so they are only estimates and thus should be used carefully. They only represent a guide as to the relative costs of different classes.
2. A weighted pop takes into account the age, sex, and ethnicity characteristics of a population, to capture the expected demand on health services.
3. Again, this is an estimate as many patients will overlap between classes, and hospitals often classify their patients in different ways.

Avondale Cost Model

1. Cost Parameters:

	GM	GS	PHA
1.1 Specialty Fixed Costs	£3,500,000	£3,850,000	n/a
1.2 Overhead Costs	£1,000,000	£940,000	n/a
1.3 Ward Cost	£0	£0	£25,000
1.4 HindleTech Costs	£75,000	£75,000	n/a
1.5 Variable Cost per Episode	£300.00	£358.70	£800
1.6 Total Number of Beds	125	110	n/a

2. Primary Clinical Inputs:

	GM	GS	PHA
2.1 Crossbay Patients	5500	4600	600
2.2 Average LOS	7.5	6.8	n/a

3. Outputs:

	GM	GS	PHA
3.1 Safe Bed Occupancy Limit	95%	95%	n/a
3.2 Maximum Episodes Possible @ max B.O.	5779	5609	n/a
3.3 Episodes Treated	5500	4600	600
3.4 Achieved Bed Occupancy (B.O.)	90.4%	77.9%	n/a
3.5 Total Variable Cost (Episode-related)	£1,650,000	£1,650,000	£480,000
3.6 Total Cost of Treatments	£6,225,000	£6,515,000	£505,000
3.7 Cost per Episode	£1,132	£1,416	£842

4. Budget Calculations:

	GM	GS	PHA
4.1 Income from CCG (price/case)	£1,154	£1,400	£775
4.2 Total Income from Crossbay CCG	£6,347,000	£6,440,000	£465,000
4.3 Total Cost of All Treatments	£6,225,000	£6,515,000	£505,000
4.4 net surpluses	£122,000	-£75,000	-£40,000
4.5 Total Income from Crossbay CCG	£13,252,000		
4.6 Total Cost of All treatments	£13,245,000		
4.7 Overall net surplus	£7,000		

5. Summary Table:

	GM	GS	PHA
5.1 Number of Patients Increased	500	0	600
5.2 Percentage Change	10.00%	0.00%	600
5.3 Change in cost per Episode	£68	-£16	
5.4 Change in income from CCG (price/case)	£46	£0	
5.5 % of Bed occupancy Left	0.0%	17.1%	
5.6 Number of Beds Left	279	1009	

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5. New Ward:

extra GM beds:	
extra GS beds:	
ward conversion cost:	

Notes

A new ward can fit up to 20 beds

Includes all new beds, equipment and IT

6. HindleTech Purchase:

HindleTech purchased?	yes
purchase cost of HindleTech:	£150,000
Hindletech B.O. limit:	95%

Notes

Input 'yes' or 'no'

Costs shared equally between GM and GS

Maximum safe limit for HindleTech

7. PHA:

ward conversion cost:	£25,000
cost per episode:	£800

Notes

Includes all new PHA equipment

Avondale Cost Model

1. Cost Parameters:

	GM	GS	PHA
1.1 Specialty Fixed Costs	£3,500,000	£3,850,000	n/a
1.2 Overhead Costs	£1,000,000	£940,000	n/a
1.3 Ward Cost	£0	£0	£25,000
1.4 HindleTech Costs	£75,000	£75,000	n/a
1.5 Variable Cost per Episode	£300.00	£358.70	£800
1.6 Total Number of Beds	125	110	n/a

2. Primary Clinical Inputs:

	GM	GS	PHA
2.1 Crossbay Patients	5400	4600	500
2.2 Average LOS	7.5	7	n/a

3. Outputs:

	GM	GS	PHA
3.1 Safe Bed Occupancy Limit	95%	95%	n/a
3.2 Maximum Episodes Possible @ max B.O.	5779	5448	n/a
3.3 Episodes Treated	5400	4600	500
3.4 Achieved Bed Occupancy (B.O.)	88.8%	80.2%	n/a
3.5 Total Variable Cost (Episode-related)	£1,620,000	£1,650,000	£400,000
3.6 Total Cost of Treatments	£6,195,000	£6,515,000	£425,000
3.7 Cost per Episode	1,147	1,416	£850

4. Budget Calculations:

	GM	GS	PHA
4.1 Income from CCG (price/case)	£1,125	£1,400	£800
4.2 Total Income from Crossbay CCG	£6,075,000	£6,440,000	£400,000
4.3 Total Cost of All Treatments	£6,195,000	£6,515,000	£425,000
4.4 net surpluses	-£120,000	-£75,000	-£25,000
4.5 Total Income from Crossbay CCG	£12,915,000		
4.6 Total Cost of All treatments	£13,135,000		
4.7 Overall net surplus	-£220,000		

5. Summary Table:

	GM	GS	PHA
5.1 Number of Patients Increased	400	0	500
5.2 Percentage Change	8.00%	0.00%	500
5.3 Change in cost per Episode	£53	-£16	
5.4 Change in income from CCG (price/case)	£75	£0	
5.5 % of Bed occupancy Left	10.0%	14.8%	
5.6 Number of Beds Left	379	848	

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5. New Ward:

extra GM beds:	
extra GS beds:	
ward conversion cost:	£200,000

Notes
A new ward can fit up to 20 beds

Includes all new beds, equipment and IT

6. HindleTech Purchase:

HindleTech purchased?	yes
purchase cost of HindleTech:	£150,000
Hindletech B.O. limit:	95%

Notes
Input 'yes' or 'no'
Costs shared equally between GM and GS
Maximum safe limit for HindleTech

7. PHA:

ward conversion cost:	£25,000
cost per episode:	£800

Notes
Includes all new PHA equipment

Avondale Cost Model

1. Cost Parameters:

	GM	GS	PHA
1.1 Specialty Fixed Costs	£3,500,000	£3,850,000	n/a
1.2 Overhead Costs	£1,000,000	£940,000	n/a
1.3 Ward Cost	£0	£0	£0
1.4 HindleTech Costs	£105,000	£0	n/a
1.5 Variable Cost per Episode	£300.00	£358.70	£0
1.6 Total Number of Beds	125	110	n/a

2. Primary Clinical Inputs:

	GM	GS	PHA
2.1 Crossbay Patients	5300	5200	0
2.2 Average LOS	7.5	6.8	n/a

3. Outputs:

	GM	GS	PHA
3.1 Safe Bed Occupancy Limit	95%	95%	n/a
3.2 Maximum Episodes Possible @ max B.O.	5779	5609	n/a
3.3 Episodes Treated	5300	5200	0
3.4 Achieved Bed Occupancy (B.O.)	87.1%	88.1%	n/a
3.5 Total Variable Cost (Episode-related)	£1,590,000	£1,865,217	£0
3.6 Total Cost of Treatments	£6,195,000	£6,655,217	£0
3.7 Cost per Episode	£1,169	£1,280	£0

4. Budget Calculations:

	GM	GS	PHA
4.1 Income from CCG (price/case)	£1,175	£1,280	
4.2 Total Income from Crossbay CCG	£6,227,500	£6,656,000	£0
4.3 Total Cost of All Treatments	£6,195,000	£6,655,217	£0
4.4 net surpluses	£32,500	£783	£0
4.5 Total Income from Crossbay CCG	£12,883,500		
4.6 Total Cost of All treatments	£12,850,217		
4.7 Overall net surplus	£33,283		

5. Summary Table:

	GM	GS	PHA
5.1 Number of Patients Increased	300	600	0
5.2 Percentage Change	6.00%	13.04%	0
5.3 Change in cost per Episode	£31	£120	
5.4 Change in income from CCG (price/case)	£25	£120	
5.5 % of Bed occupancy Left	10.0%	6.9%	
5.6 Number of Beds Left	479	409	

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5. New Ward:

extra GM beds:	
extra GS beds:	
ward conversion cost:	£200,000

6. HindleTech Purchase:

HindleTech purchased?	yes
purchase cost of HindleTech:	£105,000
Hindletech B.O. limit:	95%

7. PHA:

ward conversion cost:	£25,000
cost per episode:	£800

	GM	GS
Ward % Split		
HindleTech % Split	1.00	0.00

£187,981.33	1%
£14,098,600.00	75%
£11,278,880.00	60%
£12,688,740.00	67.50%
£13,158,693.33	70%
£12,218,786.67	65%

Notes
A new ward can fit up to 20 beds

Includes all new beds, equipment and IT

Notes
Input 'yes' or 'no'
Costs shared equally between GM and GS
Maximum safe limit for HindleTech

Notes
Includes all new PHA equipment

ccg want to be at 60% +/-

				£18,798,133.00	100%	
	GM	GS	PHA	£187,981.33	1%	
Net Spend	-£220,000.00	£7,000.00	£33,282.61			
Number of Patients	10500	10700	10500	£14,098,600.00	75%	ccg want to be at 60% +/-
Money received	£12,915,000.00	£13,252,000.00	£12,883,500.00			
% of Budget	68.70%	70.50%	68.54%	£11,278,880.00	60%	
% of Last Budget	62.29%	63.92%	62.14%			
Change last year	£475,000.00	£812,000.00	£443,500.00	£12,688,740.00	67.50%	
				£13,158,693.33	70%	
				£12,218,786.67	65%	
				£12,440,000.00	Previous Year	
				£207,333.33	£20,733,333.33	